

Static vs Contextual Embeddings for Lexical Ambiguity

Method and Results Appendix

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1 Research question and scope

This project asks whether a context-sensitive target representation distinguishes same-sense from different-sense uses more effectively than static alternatives. It compares type-level GloVe, a lexical GloVe context mean, and an unfine-tuned BERT-base target representation. It is a controlled representation probe, not a claim about fine-tuned task state of the art.

2 Dataset provenance and audit

The source is the official SuperGLUE v2 WiC archive at <https://dl.fbaipublicfiles.com/gluue/superglue/data/v2/WiC.zip>. Its SHA-256 is `ee7e67f4ae9eafbf533780faa198e62167f3cda54256cdf261877be3c0e90900`. The release is distributed under CC BY-NC 4.0 by the WiC authors.

Table 1: Validated release counts and span diagnostics.

Split	Rows	Positive	Negative	Unlabeled	Lemma mismatch	Repeated surface
Train	5,428	2,714	2,714	0	1,852	93
Validation	638	319	319	0	251	1
Test	1,400	0	0	1,400	521	20

All target spans are non-empty half-open character intervals. IDs are unique within splits. Four symmetric duplicate pairs occur within training, none within validation, and no task pair crosses train and validation. The label-hidden test split is audited but not evaluated.

3 Representations

3.1 Static baselines

GloVe uses the 6B-token, uncased, 300-dimensional release. Archive SHA-256: `617afb2fe6cb085c235baf7a465b96f4112bd7f7ccb2b2cbd649fed9cbcf2fb`. The target diagnostic looks up the supplied lemma on both sides, making every available paired cosine exactly 1 by construction. Eighteen training lemmas are OOV; missing target scores are replaced by the median finite training score.

Context vectors average in-vocabulary regex word tokens after excluding only the token span that overlaps the marked target. Candidate windows are ± 2 , ± 5 , ± 10 , and the sentence. If a context side has no usable token, it falls back to that side’s target lemma; this occurs once per window in training and zero times in validation. No evaluated context pair remains missing.

3.2 Contextual target

The encoder is `google-bert/bert-base-uncased`, revision `f584a14f45f850d60d0036d54f3e4dfdb3d52075`. It remains in evaluation mode; there is no fine-tuning. A fast tokenizer emits character offsets. Every WordPiece satisfying `token_end > target_start` and `token_start < target_end` is selected, ignoring zero-width special tokens, and selected WordPieces are averaged. Layers 1-12 and mean-last-four are cached from one forward pass.

4 Selection and evaluation protocol

The balanced training labels are stratified into 4,342 tune and 1,086 internal holdout rows using seed 2026. Candidate thresholds are fitted on tune rows by macro F1. Candidate representations are ranked on internal-holdout macro F1, then accuracy, ROC-AUC, and declared simplicity order. Winners are frozen before validation metrics are computed. Thresholds are then refitted on all 5,428 training rows. Selection ledger hash: `debfd61b068f22ea1a737f238bd90ea4fb8bf44f675907e428949b559a0bb309`.

Table 2: Training-only candidate selection.

Family	Candidate	Tune macro F1	Holdout macro F1	Selected
Static context	glove_context_2	0.5840	0.5961	yes
Static context	glove_context_5	0.5883	0.5928	no
Static context	glove_context_10	0.5917	0.5863	no
Static context	glove_context_sentence	0.5930	0.5845	no
BERT	layer_12	0.6789	0.7145	no
BERT	mean_last_four	0.6891	0.7247	yes

5 Primary validation results

Predictions use score $\geq$ threshold as same-sense. Precision and recall use same-sense as the positive label. Confusion-matrix layout is TN, FP / FN, TP. Intervals are percentile intervals from 1,000 fixed-seed paired example resamples.

Table 3: Primary results on 638 balanced validation pairs.

System	Threshold	Accuracy	Accuracy 95% CI	Macro F1	Precision	Recall	ROC-AUC
GloVe target	1.0000	0.5000	[0.4639, 0.5392]	0.3333	0.5000	1.0000	0.5000
GloVe context (± 2)	0.6506	0.5549	[0.5156, 0.5956]	0.5511	0.5670	0.4639	0.5800
BERT mean last four	0.5813	0.6708	[0.6364, 0.7085]	0.6673	0.6416	0.7743	0.7163

The selected BERT representation is correct on 428 pairs; GloVe context is correct on 354. BERT alone is correct on 178 pairs, GloVe context alone on 104, both on 250, and both primary systems are wrong on 106.

Table 4: Paired primary-system differences with 95% intervals.

Contrast	Metric	Estimate	95% interval
glove_target - glove_context_2	accuracy	-0.0549	[-0.1176, 0.0094]
glove_target - glove_context_2	macro f1	-0.2178	[-0.2661, -0.1710]
glove_target - glove_context_2	roc auc	-0.0800	[-0.1273, -0.0378]
glove_target - bert_mean_last_four	accuracy	-0.1708	[-0.2195, -0.1238]
glove_target - bert_mean_last_four	macro f1	-0.3340	[-0.3739, -0.2953]
glove_target - bert_mean_last_four	roc auc	-0.2163	[-0.2588, -0.1775]
glove_context_2 - bert_mean_last_four	accuracy	-0.1160	[-0.1693, -0.0658]
glove_context_2 - bert_mean_last_four	macro f1	-0.1161	[-0.1690, -0.0664]
glove_context_2 - bert_mean_last_four	roc auc	-0.1363	[-0.1912, -0.0811]

The accuracy, macro-F1, and ROC-AUC intervals for GloVe context minus BERT all exclude zero. This supports a difference under the current paired sample and protocol; it does not provide a universal model ordering.

6 Secondary BERT layer analysis

Each layer threshold is trained on all training pairs. Validation layer results are descriptive and do not replace the frozen primary representation.

Table 5: Individual BERT target layers.

Layer	Threshold	Accuracy	Macro F1	ROC-AUC
1	0.8546	0.5705	0.5685	0.5900
2	0.8145	0.6019	0.6014	0.6187
3	0.7590	0.6113	0.6113	0.6357
4	0.7330	0.6317	0.6314	0.6575
5	0.6949	0.6458	0.6451	0.6768
6	0.6564	0.6442	0.6419	0.6853
7	0.6299	0.6489	0.6467	0.6946
8	0.6039	0.6599	0.6558	0.7036
9	0.5728	0.6661	0.6624	0.7093
10	0.5625	0.6599	0.6543	0.7197
11	0.5947	0.6536	0.6508	0.7096
12	0.4942	0.6364	0.6308	0.6971

Accuracy rises broadly from layer 1 to layer 9 and then declines through layer 12. Layer 10 has the largest individual-layer ROC-AUC. This is compatible with the train-selected mean-last-four result, but validation was not reused for selection.

7 Error analysis

System-agreement partitions are disjoint. There are 250 both-primary-correct, 178 BERT-only-correct, 104 static-only-correct, 54 all-system-wrong, and 52 rows where both primary systems are wrong but the one-class target diagnostic is right. For BERT, error margin is absolute distance from the frozen cosine threshold. Of 210 errors, 33 are within 0.025 and 53 occupy the top error-margin quartile.

BERT recovers 77.43% of same-sense pairs and 56.74% of different-sense pairs; GloVe context recovers 46.39% and 64.58%, respectively. These are descriptive class skews. Saved high-margin examples include a same-sense anatomical *ball/balls* pair that BERT separates strongly and different-sense *engrave* and *map* pairs that it places close together. Conversely, BERT links low-overlap same-sense examples such as *rust remover/paint remover*. Fine sense granularity and constructional similarity are plausible explanations, not causal findings.

8 Reproducibility record

The locked runtime is Python 3.11.15, NumPy 2.4.6, PyTorch 2.11.0+cu130, and Transformers 5.17.0. Full inference ran on an NVIDIA GeForce RTX 3050 Ti Laptop GPU. Arrays are keyed by dataset fingerprint, model revision, maximum length, seed, and cache schema. The clean cached reproduction recorded Git commit 7c6c956a173920b8aa685e206b9e4cdade1afdb6 on branch `main`.

9 Limitations

The sample covers one English benchmark; bootstrap intervals omit dataset-design and modeling uncertainty. BERT is an older frozen base encoder. The static context baseline has no syntax, weighting, subword model, or supervised classifier. A global cosine threshold compresses potentially useful geometry. Surface-mismatch groups are observational and confounded. No score is reported for the label-hidden test split.

10 References

References

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